

Canonical Repair with Certified Numerical Releases

Closing the two remaining implementation qualifications
for counterfactual data curation

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Outcome. A new strict mode now has both *byte-canonical aggregate checkpoints* and *rigorous numerical head certificates*. It was implemented and validated locally, using the same 100 natural Civil Comments records and their original responses. The fast floating mode remains available with its previous contract; this report gives a stronger, separately specified mode.

The decisive construction is a unique rational chart of the joint range of each Gram/cross coefficient. Its pivot identity rows are implicit, eliminating the dense-versus-compact mode choice. Exact coefficient updates remove history-dependent rounding. An exact rational residual then checks a fast floating head against the exact retained-data ridge objective.

Executed evidence. All **140 deletion checkpoints** matched fresh exact retained-data rebuilds byte for byte. Ten additional final-set comparisons matched batching, singleton order, reversed order and strict reload. Every head passed the prescribed 10^{-10} tolerance; the largest rigorous Frobenius error radius was 4.336×10^{-14} . Full deletion produced empty coefficients, an exactly zero head and an exactly zero certificate bound.

Measured price. At $d = 768, c = 1$, the initial canonical checkpoint is 1,514,487 bytes, while the recursive Python live-size estimate is 11,053,348 bytes. Build took 1.869 seconds; median timed repair and certification were 0.919 and 0.638 seconds. These are distinct accounting layers and unreplicated timers, not a process-memory bound or an end-to-end speed claim.

Scope. Exactness means exact aggregation of frozen FP32 features and FP64 responses interpreted as binary rationals, with frozen curation decisions. It does not mean bit equality to legacy ordered floating-point retraining. The experiment uses lexical engineering features, not a semantic encoder or an NLP utility evaluation. No synthetic empirical documents or labels and no external computation were used.

Evidence ledger and cost

The curator uses 128-dimensional lexical hashing, cosine threshold 0.6 and a stable ID-derived priority. Learner feature dimensions are 16, 64 and 768. The seven-output panel uses all seven original toxicity fields. The regular run has horizon eight and binary64 input $\lambda = 0.01$; requests are the existing R, U and A families. Deletion checkpoints are 1, 2, 4 and 8, so timed repair batches have sizes 1, 1, 2 and 4. The full-deletion stress uses $d = 16$.

This is a convenience preview, not a representative corpus sample. The original acquisition note is included with the fixture and distinguishes connector-rendered records from original corpus bytes.

d	c	Rational entries	JSON MB	Python MB	Max bits N/D
16	1	1,865	0.048	0.317	74/72
64	1	8,705	0.189	1.069	57/57
768	1	125,570	1.514	11.053	94/92
768	7	126,566	1.529	11.120	94/92

MB means 10^6 bytes. JSON includes the canonical logical state. Python live size is a recursive estimate including rational objects, integer limbs and metadata; it excludes external inputs, process/runtime overhead, allocator slack and transient repair/serialization/native workspaces. At $d = 768, c = 1$, the numerator/denominator integer payload alone is 184,525 bytes. Neither that payload nor JSON size should be presented as actual total RAM. No exact-mode process-peak experiment is claimed here.

d	c	Build s	Median repair s	Median cert. s	Median fresh s
16	1	0.420	0.010	0.005	0.317
64	1	3.753	1.614	1.041	2.706
768	1	1.869	0.919	0.638	1.577
768	7	1.770	1.384	0.957	1.573

Repair, fresh construction and certification timers are separate. They do not include a complete service transaction; in particular some loading, serialization and numerical-candidate work are outside those timers. The certification cost can consume the apparent repair advantage. These local observations do not establish a speedup or corpus-scale throughput.

- Forty regular checkpoints span the three scalar panels and one seven-output panel. Every oracle rescored the retained graph independently, retained stable public IDs, rebuilt exact coefficients and compared all bytes.
- One hundred additional sequential deletion checkpoints also compared fresh exact bytes and checked certificates. The last result is exactly empty.
- Ten comparisons exercised batch/singleton/reversed histories and strict reload. Same-runtime numerical heads also agreed; portable floating-head bytes are not part of the theorem.
- Duplicate, unknown, over-budget and retried requests were rejected without changing input bytes. The service additionally accepted the required tolerance and rejected an unattainable zero tolerance without release.

All five central source hashes in the run ledger and the natural-data hash were independently rechecked after execution. The archive additionally hashes the curator, dependency helpers and compiled native binary.

Algorithm and release boundary

For each current coefficient, the strict implementation stores its exact joint-range chart, pivot Gram/cross cores and count. The authoritative state is one canonical JSON byte string containing sorted keys, normalized rational numbers, sorted alive IDs, fixed dimensions, public universe and remaining horizon. Solver output, request history and previous factor charts are absent.

1. **Initialize once.** Freeze the numerical inputs, build exact coefficient endpoints in blocks of at most $\min(d, 128)$ contributors, and canonicalize each block/merge. No family of dense ambient Grams is built first.
2. **Repair.** Validate the request; substitute the deleted IDs into keys, prune degrees above the decreased horizon, merge collisions in rational arithmetic, and remove exactly zero coefficients. Commit only after the staged new state succeeds. The current reference scans/sorts all keys.
3. **Propose.** Reconstruct the current coefficient's rational basis, convert for a numerical QR-based reduced solve, and obtain a finite candidate head. The rational basis is not orthonormal, so the old reduced equation cannot be applied directly.
4. **Certify.** Prove the current Gram core is PSD by an exact rational Schur-complement test and evaluate the full candidate residual exactly. Check the requested error tolerance as a rational inequality.
5. **Release.** Return canonical updated bytes, the candidate head and its certificate only if the check passes. The pure bytes-in/bytes-out service leaves its input unchanged on failure.

The certificate covers numerical fidelity to the supplied exact target. A valid retained-data target is established by the repair theorem and verified against fresh rebuilds in this run. A structurally canonical but maliciously altered state is not automatically authentic. Dataset/configuration hashes and provenance belong in the surrounding execution manifest.

Independent checks. The numerical certificate passed 243 exact rational oracle comparisons and 1,984 upward-square-root checks. The chart passed 66 independent ambient dense-RREF comparisons plus three large-integer cases. Twenty native/Python algebra fixtures and 50 certificate module checks also passed. Six malformed checkpoint variants reject even under `python -O`; that audit found and fixed a Boolean/integer alias before the frozen full run. These small algebra fixtures are software tests, not synthetic empirical datasets.

Implementation trust and portability. Fast exact canonicalization uses the installed GMP 6.3.0 runtime through a local Linux LP64 helper. Its explicit ABI declarations have static checks, startup arithmetic checks and independent large-integer comparisons. Source and binary are included. An exact Python `Fraction` fallback avoids that platform dependency at additional runtime cost. Both produce the same canonical representation.

1 Fixed-blocker curation and the repair polynomial

This note concerns a specified fixed-blocker selector. For record v , let $o(v)$ be its deletion-unit owner and let $B(v)$ be its distinct blocker-owner set. A record is selected precisely when its owner survives and all its blocker owners are deleted. In the engineering curator, blockers are all earlier priority neighbors above the frozen similarity threshold. This is not a connected-component representative rule or a universal description of semantic deduplicators.

Let $t_u \in \{0, 1\}$ indicate deletion of unit u . With frozen feature z_v and response y_v , write

$$s_v = (z_v z_v^\top, z_v y_v^\top, 1), \quad P(t) = \sum_v s_v (1 - t_{o(v)}) \prod_{u \in B(v)} t_u.$$

If $o(v) \in B(v)$, its Boolean contribution is zero. Otherwise each record has a positive coefficient endpoint $B(v)$ and a negative endpoint $B(v) \cup \{o(v)\}$. At horizon h , retain endpoints of degree at most h . For every request of at most h deletions, all omitted monomials evaluate to zero, so this truncated polynomial gives exact selected-set moments.

After a request F , substitute $t_u = 1$ for $u \in F$, giving the key map $J \mapsto J \setminus F$, and reduce the horizon to $h - |F|$. A discarded degree- k term has new degree at least $k - |F| > h - |F|$ and cannot reenter. A record whose owner is deleted has identical positive and negative endpoints, which cancel. A surviving record has blocker set $B(v) \setminus F$. Consequently repaired coefficients equal fresh retained-data coefficients.

Define E_h as the number of current records with own owner absent from their blocker set and at most h distinct blocker owners. Let m_J count nonzero record contributions incident to coefficient J . Then

$$K \leq 2E_h, \quad \sum_J m_J \leq 2E_h, \quad \text{rank}([G_J \ H_J]) \leq m_J.$$

These are the rank and key bounds used below. They remain true for repeated source owners, signed cancellations, horizon zero and an empty state. In a fresh retained rebuild, blocker row positions must be remapped while original public owner IDs and the public universe remain fixed.

An aggregate can encode recoverable facts about individual records. The contract is absence of a separate payload table and of repair-time raw-row reads, not cryptographic privacy. Similarly, semantic curation itself requires separate empirical validation. The proofs below hold for the stated selector.

2 Exact canonical repair with a rigorous numerical release certificate

Numerical target. Interpret each frozen finite FP32 feature coordinate and FP64 response coordinate as its exact dyadic rational value. The graph, priority order, source ownership, and curation threshold decisions are already frozen. All coefficient aggregation and repair in this section use exact rational arithmetic. This specifies a new exact-aggregation implementation of the same mathematical learner. Byte-wise equality is to a fresh construction under this specification; it is not a claim of byte-wise equality to the previous implementation’s ordered FP64 summation and Cholesky routine. The latter remains a separate numerical comparison.

2.1 A unique rational coordinate chart

For a current coefficient let

$$M = [G \ H], \quad G = G^\top \in \mathbb{Q}^{d \times d}, \quad H \in \mathbb{Q}^{d \times c}, \quad \rho = \text{rank } M.$$

Compute the unique reduced row echelon form of $M^\top$ over $\mathbb{Q}$, remove its zero rows, and transpose. Call the resulting $d \times \rho$ matrix U . Let $P = (p_1, \dots, p_\rho)$ be its ascending pivot row indices. Then $U[P, :] = I_\rho$. Define

$$A = G[P, P], \quad B = H[P, :], \quad F = U[P^c, :].$$

Store (P, F, A, B, n) , packing the symmetric A . The identity pivot rows of U are implicit and are never stored. For $\rho = 0$ only the count is needed; a key is removed precisely when $\rho = 0$ and $n = 0$. Every rational is stored in lowest terms with a positive denominator.

Theorem 2.1 (Unique exact coefficient encoding). *The tuple (P, F, A, B, n) is a function only of the exact current coefficient (G, H, n) and the fixed ambient coordinate order. It reconstructs*

$$G = UAU^\top, \quad H = UB.$$

In particular, it is independent of contributor ordering, prior bases, collision order, and whether an earlier representation happened to be dense.

Proof. Reduced row echelon form uniquely identifies a row space. Hence U and P depend only on the joint column space of G and H . Every column of both lies in $\text{range}(U)$; because G is symmetric, its row space lies there too. Thus $G = UCU^\top$ for some C , and restricting both coordinates to P gives $C = G[P, P] = A$. Likewise restricting $H = UD$ to P gives $D = B$. Canonical rational normalization removes numerator and denominator representation choices. $\square$

Theorem 2.2 (Current-eligibility entry bound without a mode bit). *Let E_h be the number of currently eligible records at remaining horizon h . Across all live keys, the number of stored rational coefficient entries, including counts, is at most*

$$\sum_J \left[(d - \rho_J)\rho_J + \frac{\rho_J(\rho_J + 1)}{2} + c\rho_J + 1 \right] \leq 2E_h(d + c + 1).$$

There are at most $2E_h$ pivot indices, at most $2E_h$ live keys, and $O(hE_h)$ key incidences. A public alive-unit registry adds $O(U)$ identifiers. For every individual rank, the coefficient entry count excluding its count is no greater than the packed dense Gram/cross count.

Proof. As before, each eligible record contributes to at most two endpoints and $\sum_J \rho_J \leq 2E_h$. For integer $\rho \geq 0$,

$$(d - \rho)\rho + \rho(\rho + 1)/2 + c\rho = (d + c)\rho - \rho(\rho - 1)/2 \leq (d + c)\rho.$$

Add at most $2E_h$ counts. The pivot index sum is $\sum_J \rho_J$. The displayed quadratic expression is nondecreasing on integer $0 \leq \rho \leq d$ and at $\rho = d$ equals $d(d+1)/2 + dc$. At full rank $U = I_d$ is entirely implicit. Therefore no history-dependent dense/compact choice is necessary. $\square$

The theorem counts rational entries, not eight-byte floating-point words. It neither promises that exact rational storage is smaller in bytes than FP64 factors nor transfers a measured floating-point compression ratio.

2.2 Exact merge without ambient Gram materialization

For two entries reconstruct their thin bases U_1, U_2 from their free rows and implicit pivot identities, and put

$$L = [U_1 \ U_2], \quad D = \text{diag}(A_1, A_2), \quad E = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}.$$

The sum is $(LDL^\top, LE, n_1 + n_2)$. Canonicalize the column space of L to a full-column-rank chart V with pivot rows P_V . With $T = L[P_V, :]$, one has $L = VT$. Form only the small matrices

$$K = TDT^\top, \quad Q = TE.$$

Let R span the column space of $[K \ Q]$. Canonicalize the column space of VR to the final U , with pivot rows P . The final cores can be computed directly from rows of L :

$$A = L[P, :]DL[P, :]^\top, \quad B = L[P, :]E.$$

All ranks and zero tests are exact. The intermediate basis choices do not affect the final canonical chart. For colliding total width $s = \rho_1 + \rho_2$, a conventional elimination implementation uses $O(ds + s^2 + sc)$ rational workspace and $O(ds^2 + s^3 + s^2c)$ rational arithmetic operations. This excludes integer bit complexity. Processing one collision at a time avoids any persistent Kd^2 array family. Construction must likewise stream or aggregate into bounded per-key workspaces, not first construct the old dense state.

Theorem 2.3 (Byte-canonical history-independent repair). *Fix the initial universe and frozen configuration, the final retained records, and the remaining horizon. Exact coefficient substitution, exact collision merging, degree pruning, and zero-key pruning produce the same canonical coefficient state as fresh construction from those retained records. Its serialization is identical under a fixed canonical format.*

Proof. Exact substitution and pruning obey the finite-horizon polynomial identity, hence produce the same decoded coefficient map as fresh construction. The preceding theorem gives a unique encoding of each coefficient. Sorting keys and alive public identifiers, normalizing each rational, and fixing metadata and byte encoding therefore give identical bytes. The remaining horizon is included in the state and must agree. $\square$

A canonical format must not include request logs, previous bases, insertion order, allocator identity, pickle object IDs, nondeterministic compression metadata, timestamps, or a floating solver result. Configuration hashes, unit-order conventions, dimensions, remaining horizon, sorted alive-unit IDs, sorted keys, pivot indices, and normalized exact numbers suffice. If numerical releases are retained as state, they require their own canonical numerical specification or must be excluded from this theorem. The contract concerns reachable current logical state, not erasure of allocator remnants, backups, or previously released models.

Implemented repair work and staging. The current exact implementation sorts and scans every live coefficient key; it does not yet use its reconstructed inverse and degree indexes to restrict work to touched keys. For K keys of degree at most h , charge $O(K(h+1)\log(K+1))$ comparison and key-processing work in addition to the exact collision arithmetic above. The harmless $+1$ terms cover empty keys, $h=0$, and $K=0$. An update stages its new coefficient map before committing, while the old live map remains available for atomic failure. Unchanged factor objects can be shared, but changed factors and old factors can coexist temporarily. This transient old/new-state overlap, canonical serialization copies, and exact-number temporaries are outside the persistent entry bound and must be included in measured peak memory.

2.3 Bit complexity and the price of exactness

Because the inputs are finite binary floats, every exact coefficient entry has a finite common power-of-two denominator. For a nonzero coefficient, choose $Q \geq 0$ such that $2^Q[G\ H]$ is integral. Define L to bound both $Q+1$ and the signed bit length of every entry of this integer matrix. Thus L accounts for input exponents as well as summation growth; merely bounding scaled integer numerators would not suffice for tiny dyadics. Use the maximum such L across live coefficients; when no coefficient is live the numerical-storage bound is zero without defining L . At rank ρ , every free entry of the canonical chart is a ratio of minors of order at most ρ . By Hadamard's inequality, its numerator and denominator have

$$O(\rho(L + \log(\rho + 1)))$$

bits. The cores A and B are submatrices of the exact moments and retain their dyadic form. Thus a conservative uniform persistent bit bound is

$$O(E_h(d+c)d(L + \log(d+1)))$$

plus the key and public-identifier encoding. This is deliberately a worst case; actual bit lengths, serialized bytes, process peak memory, and time must be measured. The loss of the Kd^2 *entry-count* term does not imply fixed-width rational entries or a universal physical-byte win.

Use fraction-free elimination or normalized rational arithmetic with exact zero pivots; numerical rank thresholds are not part of this mode. The rational-operation work bound above must be multiplied by appropriate integer-arithmetic costs at the encountered bit lengths. Final rational normalization controls persistent representation; temporary numerator growth must still be charged. It can make an exact mode substantially slower than the floating mode. This section supplies a correctness construction, not a claim that exact elimination is a competitive high-throughput kernel.

2.4 Rigorous certification of a floating release

Let (G, H, n) be the exact current selected-set coefficient, with $G \succeq 0$ and $n > 0$. Interpret the positive regularization input λ exactly as a rational and write $\alpha = \lambda n$. The exact mathematical ridge head is

$$W_* = (G + \alpha I)^{-1}H.$$

The rational chart is not orthonormal. A reduced exact equation is

$$W_* = UZ, \quad (A(U^\top U) + \alpha I)Z = B.$$

The older equation $(A + \alpha I)Z = B$ is valid only for an orthonormal basis and must not be reused for this chart. An approximate head may be obtained by any finite-precision solver, including a QR-transformed small system. Let its returned finite binary-float matrix be $\widehat{W}$, itself interpreted exactly as a dyadic matrix for certification.

Theorem 2.4 (Exact-residual roundoff certificate). *Compute the exact rational residual and its squared norm by*

$$R = U \left[B - A(U^\top \widehat{W}) \right] - \alpha \widehat{W}, \quad S = \sum_{i,j} R_{ij}^2.$$

Then

$$\|\widehat{W} - W_*\|_F^2 \leq \frac{S}{\alpha^2}.$$

This bound includes all floating error in the solver and coordinate transformations that produced $\widehat{W}$. It requires no ambient $d \times d$ Gram allocation and no floating backward-error assumption.

Proof. The residual is exactly $H - (G + \alpha I)\widehat{W}$. Since $G \succeq 0$, the smallest eigenvalue of $G + \alpha I$ is at least α . Therefore $\widehat{W} - W_* = -(G + \alpha I)^{-1}R$ and $\|\widehat{W} - W_*\|_F \leq \|R\|_F/\alpha$. All arithmetic used in R and S is exact, so this is a certified bound, not merely a floating residual diagnostic. $\square$

The PSD premise follows from the valid exact selected-set construction; it is not a promise for arbitrary signed keys or an untrusted corrupted state. Counts remain exact integers. At $n = 0$, use the prescribed zero head exactly and report the empty-target case separately rather than divide by zero. If a solver returns nonfinite values, issue no certificate and fall back or fail explicitly.

The exact fraction S/α^2 is itself a machine-checkable squared error bound. To produce a decimal or binary upper radius, use an upward-rounded square root. For a rational $T = a/b \geq 0$ and chosen precision p , compute an integer k such that

$$k^2 b \geq a 2^{2p}, \quad (k-1)^2 b < a 2^{2p}$$

with integer square roots and exact comparisons. Then $k 2^{-p} \geq \sqrt{T}$. A subsequent binary64 conversion must also be rounded upward, or the exact rational radius should be exposed directly. Ordinary nearest-rounded `sqrt` and a printed floating residual are insufficient.

For any query vector z with certified $\|z\|_2 \leq R_z$, this yields $\|z^\top(\widehat{W} - W_*)\|_2 \leq R_z \sqrt{S}/\alpha$. The certificate bounds numerical fidelity to the exact mathematical ridge head. It does not certify that upstream semantic curation is accurate, that the feature map forgot its own training data, or that the polynomial coefficients conceal individual records.

2.5 Migration cannot recover discarded precision

Canonicalizing an existing floating summary can canonically encode the exact dyadic values of that *approximate* summary. It cannot in general recover the exact original-input aggregate. Floating aggregation is many-to-one: in binary64, both $1 + 0$ and $1 + 2^{-54}$ round to 1 under round-to-nearest. Such ambiguity can occur in a cross-moment sum with the same Gram and count. A procedure given only the rounded aggregate cannot know which exact target preceded it.

Consequently the exact mode must be built from authorized retained inputs, or from previously maintained exact sufficient state. If only the old floating summary survives, the strongest justified migration is an exact encoding of that approximate state plus a separately proved bound on its initial moment error. A fresh exact rebuild in an authorized local workspace is a one-time cost and must be disclosed. Subsequent exact repairs require only the exact current coefficient state and public request metadata; they do not reread the original rows.

Paper claim and reproducibility

The supportable claim is: *for fixed-blocker curation with frozen binary features and ridge sufficient statistics, there is an exact finite-horizon coefficient repair representation with unique history-independent checkpoint bytes, linear current-eligibility rational-entry count, and an exact-residual certificate for approximate numerical releases.* The bit-growth and scan/merge work bounds are explicit. The present implementation is a constructive correctness result, accompanied by natural-text engineering checks.

The chart and residual/coercivity argument use classical algebra and verified numerics. Novelty must be assessed for the curation-aware combination, deletion semantics and empirical consequence; it should not be attributed to RREF, rational arithmetic or residual bounds in isolation. For classical verified numerics, see S. M. Rump, *Verification methods: rigorous results using floating-point arithmetic*, Acta Numerica 19 (2010), 287–449, [doi:10.1017/S096249291000005X](https://doi.org/10.1017/S096249291000005X). The native helper documents the consulted official GMP interface references.

The companion archive includes executable code, the CC0 natural-data fixture, all strict-mode result records, exact checkpoint and release certificate, independent audit reports, native code, pinned requirements, complete report source and integrity manifest. The strict-mode README gives the replay commands. Replay requires no network and no external compute. Prior floating-mode results are not overwritten by this report.

Before rerunning, preserve `canonical_results/`, because the harness replaces those output records. From the extracted `empirical_execution` directory run:

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1
MKL_NUM_THREADS=1 python3 run_canonical_validation.py
```

The environment assignments belong on the same shell command line (or may be exported first). The algebra and certificate audit scripts can be run independently. The included complete L^AT_EX source can be rendered using standard PDFLaTeX with its declared packages; the builder also supports the already initialized local TeX format used for this execution.